

Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering
Course Code: CSE 3210, Course Title: Digital Signal Processing Sessional
Session:2021-2022, Lab Final, Full Marks: 15, Time: 50 minutes

CO Code	Course Outcome Statement	Mapped PO	Relevant KPA's	Bloom's Taxonomy Level
CO1	Apply mathematical knowledge to implement and analyze various signal processing systems, enhancing understanding of their principles and applications.	PO2	K2,K3,P1,P3	C3-Apply

A continuous-time biomedical signal is given as:

$$x_a(t)=3\sin(30\pi t)+4\sin(14\pi t)$$

1. Using Python demonstrate the following:
- Select a proper sampling frequency yourself such that no aliasing occurs. Sample the signal for an appropriate finite time duration and generate the discrete-time signal $x[n]$.
 - Choose a quantization step size of your preference and perform uniform quantization on the sampled signal.
 - Plot the original continuous-time signal, the sampled signal, and the quantized signal

2. Using your quantized sequence $x_q[n]$, construct a single hybrid transformed signal defined as:

$$z[n]=5*x_q(-2n-4)$$

Visually demonstrate each transformation inside the hybrid expression through proper intermediate plots.

3. A digital filter with impulse response $h[n] = [2, -1, 0, -1, 5]$ is applied to your quantized signal. Plot the convolved signal.

4. Perform a similarity analysis between the quantized signal and the filtered output manually. Based on the maximum correlation point, mention whether the filtered output is delayed, advanced, or aligned with respect to the input.

Rubrics Sample:

Criteria	Excellent	Satisfactory	Weak
Sampling & Quantization (4 marks)	Correct sampling frequency ensuring no aliasing; accurate sampled and quantized signals; plots are clear, well-labeled, and properly presented. (4)	Sampling frequency acceptable but justification/plots partially correct; minor issues in quantization or visualization. (2–3)	Incorrect sampling frequency or aliasing present; quantization wrong or missing; plots unclear or absent. (0–1)
Hybrid Transformation (4 marks)	All transformations (reversal, scaling, shifting) correctly implemented; intermediate and final plots are clear; observations correctly explained. (4)	Transformations mostly correct but with minor indexing errors; plots partly clear; limited or partially correct interpretation. (2–3)	Major mistakes in transformations; incorrect indexing; missing or unclear plots; no meaningful interpretation. (0–1)
Manual Convolution (4 marks)	Manual loop-based convolution correct; output sequence accurate; convolved output plot clear and properly labeled. (4)	Convolution partly correct with minor errors; plot present but may have labeling or scaling issues. (2–3)	Incorrect convolution logic; major indexing mistakes; missing or unreadable plot. (0–1)
Correlation & Similarity Analysis (3 marks)	Manual cross-correlation correctly implemented; correlation plot clear; correct identification of delay/advance/alignment. (3)	Correlation partially correct; plot present but interpretation partially accurate. (1–2)	Incorrect or incomplete correlation; missing plot; wrong or no interpretation of alignment. (0)

Justification for KPA:

Code	Descriptor	How It Appears in This Question	Justification
K2	Conceptually based mathematics and numerical analysis applicable to the discipline.	Students apply sampling criteria, quantization, discrete-time indexing, and numerical implementation of DSP operations using Python.	The question requires mathematical reasoning and numerical computation to model and interpret digital signals, aligning with the expectations of K2.
K3	Theory-based formulation of engineering fundamentals in the discipline.	The tasks involve core DSP fundamentals such as sampling, quantization, signal transformation, convolution, and correlation.	Students rely on essential signal processing principles and LTI system knowledge, demonstrating engagement with engineering fundamentals under K3.
P1	Requires in-depth engineering knowledge and first-principles analytical approach.	Each task depends on understanding sampling rules, discrete-time operations, system response, and similarity measures.	The problem cannot be solved without applying foundational DSP knowledge, fulfilling the depth and analytical requirements of P1.

P3	Requires abstract thinking and original analysis to model and interpret problems.	Students transform signals, implement operations without built-ins, and interpret outcomes through visual and computational analysis.	The question demands abstraction and formulation of discrete-time models, satisfying the characteristics of P3.
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